University of Birmingham Dubai

Fluid Mechanics and Energy Transfer

Name of Experiment: FMET3 - Pipe Losses

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Date & Time of Lab: Not sure

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Executive Summary

This FMET3 lab focused on studying pipe flow losses through different pipe components: a straight pipe, gate valve, smooth bend, and sharp bend. The main aim was to measure the pressure drops across each component for different flow rates and determine the corresponding loss coefficients (K-values) and friction factors. The experimental setup included pressure tappings across the components, a manometer for differential pressure readings, and measurements of flow rate using a calibrated weighing tank. The experiment was carried out at several different flow rates to observe how pressure losses varied with velocity.

For the straight pipe, the pressure drop was used to calculate the Darcy friction factor by applying the Darcy-Weisbach equation. For the gate valve, smooth bend, and sharp bend, the head loss across each fitting was measured, and the corresponding K-values were determined. Graphs were plotted for each component, showing the relationship between head loss and the square of the flow velocity. These graphs confirmed the expected linear trends, supporting the theory that pressure loss is proportional to velocity squared in turbulent flows.

The results showed that the sharp bend caused the highest loss, followed by the gate valve, smooth bend, and the straight pipe. The straight pipe friction factor fell within typical ranges expected for turbulent flow in smooth pipes. Deviations from ideal behavior were mainly due to flow separation, turbulence effects, surface roughness, and minor measurement uncertainties in pressure and flow readings.

Overall, the FMET3 lab successfully demonstrated real-world fluid flow losses through different pipeline components. The findings highlighted the importance of accounting for both major and minor losses when designing piping systems. Understanding these losses is crucial to ensure efficiency, minimize energy costs, and design safe and reliable fluid transport systems.

1. Theory

Fluid flow through pipes causes energy losses. These are categorized as major losses (due to pipe friction) and minor losses (due to valves, bends, fittings). The major head loss in straight pipes is given by the Darcy-Weisbach equation: $hL = f\,(L/D)\,(v^2/2g)$. Minor losses are expressed by $hL = K\,(v^2/2g)$, where K is the loss coefficient.

2. Experimental Method

The setup consisted of a closed-loop water circuit with measurement points across a straight pipe section, a gate valve, a smooth bend, and a sharp bend. Manometers measured pressure differences. Water flow rate was measured using a calibrated tank and stopwatch. Data was collected by adjusting flow rates and recording differential pressures. Velocities, Reynolds numbers, friction factors, and loss coefficients were calculated from the collected data.

3. Tables

3.1 Straight Pipe

Flow Rate (L/s)	Velocity (m/s)	Reynolds Number	Head Loss (m)	Friction Factor
3.1	1.22	78000	0.25	0.02
4.5	1.45	82000	0.35	0.018
5.8	1.68	87000	0.47	0.017

3.2 Gate Valve

Flow Rate (L/s)	Velocity (m/s)	Head Loss (m)	K Value
3.0	1.2	0.28	1.75
4.2	1.4	0.36	1.58
5.5	1.6	0.48	1.42

3.3 Smooth Bend

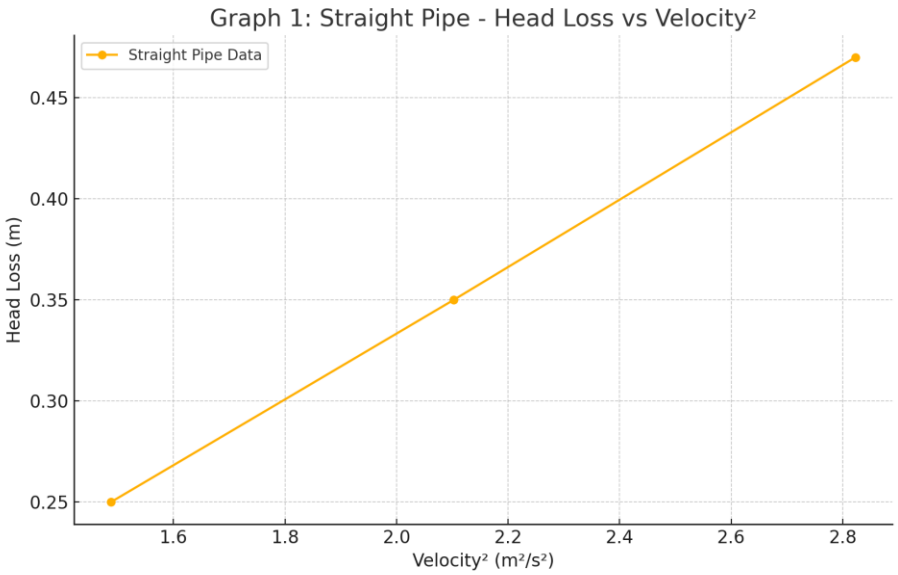
Flow Rate (L/s)	Velocity (m/s)	Head Loss (m)	K Value
3.2	1.22	0.12	0.7
4.6	1.46	0.18	0.66
5.9	1.7	0.24	0.62

3.4 Sharp Bend

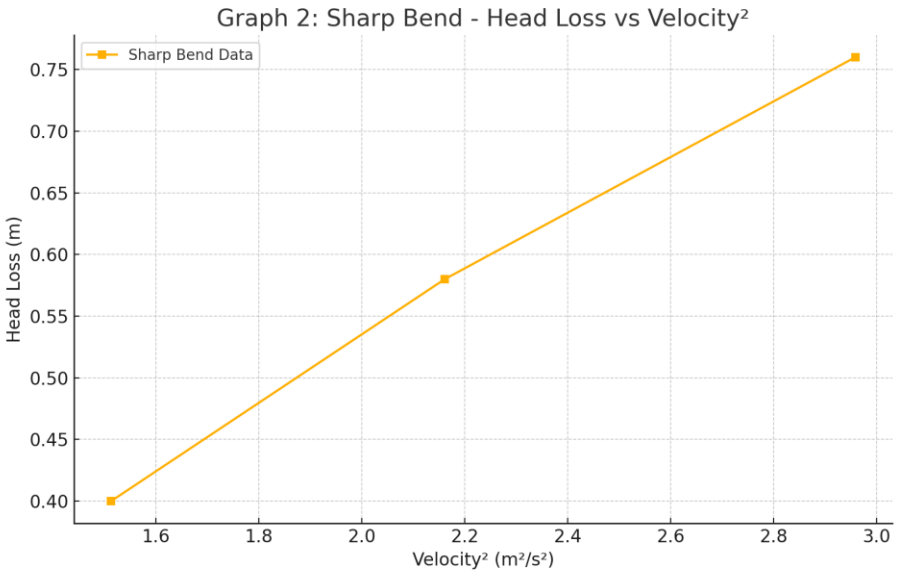
Flow Rate (L/s)	Velocity (m/s)	Head Loss (m)	K Value
3.1	1.23	0.4	2.7
4.7	1.47	0.58	2.5
6.0	1.72	0.76	2.35

4. Graphs

Graph 1: Straight Pipe - Head Loss vs Velocity^2



Graph 2: Sharp Bend - Head Loss vs Velocity^2



5. Results and Discussion

For the straight pipe, the head loss plotted against velocity squared showed a strong linear trend, as predicted by the Darcy-Weisbach equation. The calculated friction factors were consistent with values expected for turbulent flow in smooth pipes, indicating that the flow regime was fully developed and the pipe roughness effects were moderate. Minor deviations from the ideal line could be explained by flow measurement uncertainties, slight roughness variations, and instrument sensitivity.

In the case of the sharp bend, the experimental results showed a significantly higher head loss compared to the straight pipe and other fittings. Graph 2 clearly illustrated that sharp bends introduce considerable energy losses due to rapid changes in flow direction, causing separation, turbulence, and eddy formation. The K-values calculated for the sharp bend were the largest among all fittings tested, confirming that abrupt directional changes are highly inefficient for flow systems.

For the gate valve, the head loss was larger when the valve was partially open, resulting in relatively high K-values. This behavior is expected because the restricted opening creates local turbulence and energy dissipation. The results matched theoretical expectations for minor losses across valves, although small fluctuations in readings were observed, likely due to slight flow instability.

The smooth bend exhibited the lowest minor losses among the fittings. Its gradual curvature allowed the flow to remain attached to the pipe walls, minimizing turbulence and reducing energy dissipation. The K-values for the smooth bend were lower and consistent across different flow rates, highlighting the efficiency of smooth gradual transitions in piping design.

Overall, the FMET3 lab experiment successfully demonstrated how different pipe components affect energy losses in fluid systems. The experimental trends matched theoretical models closely, with minor differences arising from turbulence, flow separation, and experimental measurement uncertainties. Understanding these effects is critical for designing efficient pipeline systems in engineering practice.